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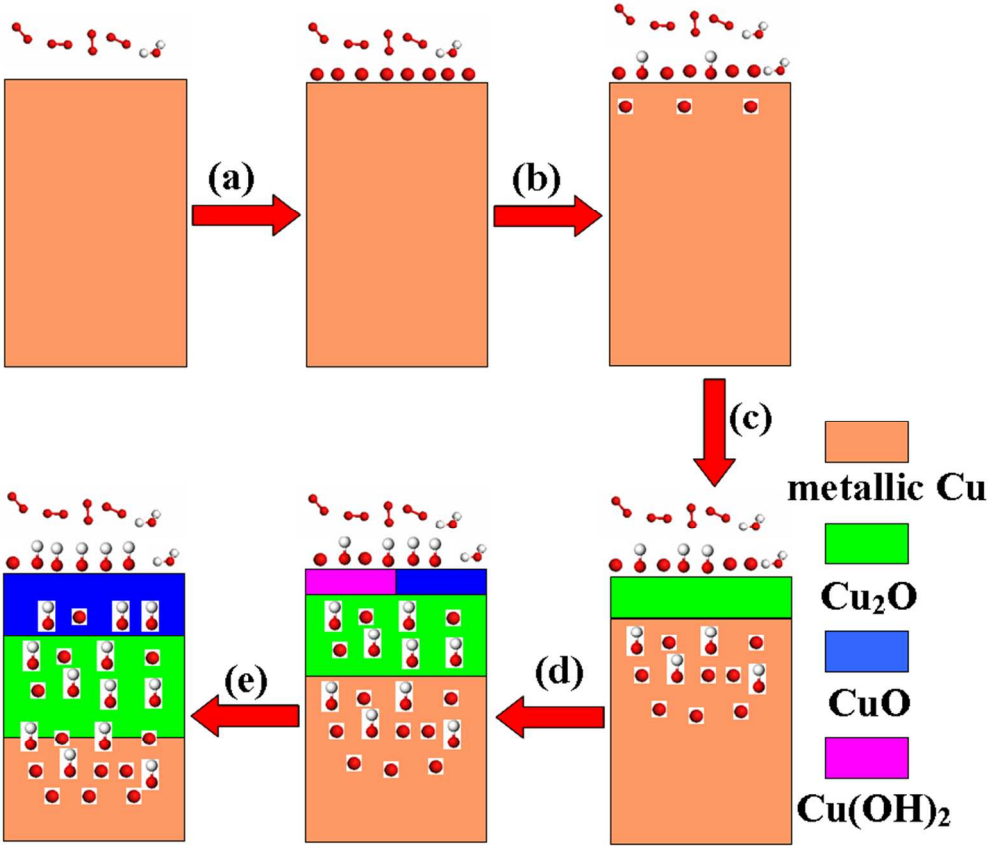
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XPS and DFT studies on the autoxidation process of Cu sheet at room temperature

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Abstract: To better understand the autoxidation mechanism of Cu-based catalysts, the oxidation of Cu sheet exposed to ultra-high vacuum and air at ambient temperature are studied using X-ray photoelectron spectroscopy (XPS) and density functional theory. Six main stages of Cu autoxidation are revealed: (1) dissociative adsorption of O₂, (2) coexistence of non-dissociative and dissociative H₂O adsorption, (3) diffusion of O and OH into Cu bulk, (4) formation of metastable Cu₂O layer, (5) further oxidation and formation of metastable Cu(OH)₂ and CuO layer, and (6) transformation phase of the metastable Cu(OH)₂ to CuO. The formation of Cu(OH)₂ depends on the relative humidity of air and the concentration of adsorbed OH of the Cu sheet. Based on these results, we propose that the preservation time of the Cu-based catalysts should be decreased or the catalysts should be preserved in a high vacuum and at low relative humidity. Considering the time of the sample preparation before ex-situ XPS analysis and other surface characterizations, the Cu-based catalysts need to be etched by about 10 s using an Ar ion gun. These findings serve as a guide for the preservation and preparation of Cu-based catalysts before actual measurement.

Keywords: density-functional calculations, X-ray photoelectron spectroscopy, Cu, autoxidation

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1. Introduction

Cu-based catalysts are used for numerous catalyst reactions, such as CO₂ or CO hydrogenation,¹⁻³ the water–gas shift reaction,^{4,5} decomposition of nitrogen oxides,^{6,7} and so on.⁸ Despite a number of mechanistic studies, the active center of Cu-based catalysts has been a matter of debate in literature over the past 20 years. Based on their opinions on active species of Cu, for example, three groups of methanol synthesis from CO or CO₂ hydrogenation were defined,⁹⁻¹¹ namely, Cu⁰, Cu⁺, and Cu⁰-Cu⁺, using X-ray photoelectron spectroscopy (XPS), infrared (IR) spectroscopy, or other analyses. Previous studies have reported the easy autoxidation of metallic surface in air;¹²⁻¹⁵ hence, the autoxidation of Cu-based catalysts can be easily performed during sample preparation before measurement. However, the valence of Cu species obtained by XPS or the adsorption behavior obtained by IR may give inaccurate results.

Given the broad range of applications including catalysis, corrosion and corrosion protection, adhesion, and microelectronics, the interaction of copper surfaces with gaseous oxygen has motivated numerous model studies.¹⁶⁻²⁴ According to these studies,^{17-18,20} the surface oxidation of Cu can be summarized in three distinct steps: first, the dissociative adsorption of O₂ on Cu surface; second, the formation of Cu₂O layer and the appearance of high elongated islands; third and last, the formation of high corrugated CuO islands. However, these studies mainly focus on the influence of temperature in the formation of Cu oxidation. For example, Lampimäki et al.¹⁸ found that both Cu₂O and CuO formations are enhanced by increased surface temperature, but no pressure dependence can be seen.

Few studies have characterized Cu sheet oxidation at room temperature conditions.¹²⁻¹⁵ However, the oxidation process of metallic Cu has been a matter of debate in literature. For example, previous studies proposed that the formation of CuO layer happens only after the complete growth of the Cu₂O layer,¹⁸ whereas recently Platzman et al.¹² considered that the formation of CuO layer occurs through the metastable formation of Cu(OH)₂ layer. Moreover, Platzman et al.¹² found that Cu²⁺ species are formed after 24 h of exposure, which is obviously shorter than previous results. The formation of Cu(OH)₂ is due to the existence of H₂O, but the role of H₂O during autoxidation remains unknown.

The aforementioned studies reported the easy autoxidation of metallic surface in the presence of

air; hence, the autoxidation of Cu-based catalysts can be easily performed during the process of catalyst preservation and preparation before measurement. Therefore, a study on the autoxidation mechanism of Cu-based catalysts at a microscopic level is urgently needed. To better understand the autoxidation process, the oxidation behavior and growth mechanism of Cu sheet exposed to ultra-high vacuum (UHV) and air at room temperature were studied in this work. Moreover, calculations based on density functional theory (DFT) were used to study the different roles of H₂O and O₂ in the Cu autoxidation process. The results can serve as a guide for the preservation and preparation of Cu-based catalysts before measurement.

2. Experimental and theoretical methods

2.1. Experimental studies

2.1.1. XPS

XPS analyses were performed using a V.G. Scientific ESCALAB250 with focused monochromated Al K α ($h\nu = 1486.6$ eV; 150 W; approximately 500 μ m diameter of irradiated area). A hemispherical electrostatic analyzer and a Large Area XL Lens were used to maximize the signal. The residual pressure inside the analysis chamber was set to less than 2.0×10^{-9} mbar. The take-off angle (θ) between the direction of the analyzer and the Cu sheet was 90°. No smoothing routine was applied to data during the analysis using the Advantage 4.84 version. A Shirley-type background subtraction was done prior to curve fitting, and the experimental curve fitting was set to 80% Gaussian to 20% Lorentzian function. In addition, a minimum number of doublets were used to fit the experimental curves.

The copper surface oxide layer thickness (d_{ox} , nm) was given by equation²⁵

$$d_{ox}(nm) = \lambda_o \sin \theta \ln \left(\frac{N_m \lambda_m I_o}{N_o \lambda_o I_m} + 1 \right)$$

where I , N , λ , d , and θ are the intensity of the photoelectron peak (i.e., peak areas), the volume density of metal atoms, the inelastic mean free path (IMFP) of photoelectrons, the oxide thickness, and the electron take-off angle with respect to the sample surface, respectively. The subscripts m and o stand for metal and oxide, respectively. The ratios of the volume density of copper atoms in bulk to oxide (N_m/N_o) are 3.28 for Cu₂O and 1.75 for CuO. The IMFP values of Cu₂O, CuO, and Cu were 2.86, 2.80, and 1.35 nm, respectively.¹² To overcome the complications associated with

the small differences in $\text{Cu}(\text{OH})_2$ and CuO , the influence of the $\text{Cu}(\text{OH})_2$ layer was neglected because of the relatively small thickness.¹² The difference in transmission functions of Cu 2p3 and Cu LMM was corrected using a correction factor (ϕ) by neglecting backscattering effects. ϕ was calculated from the metallic Cu intensity ratios of Cu 2p3 and Cu LMM spectra (the peak of No. 2, Fig. 1c).

$$\phi = \frac{I(\text{Cu}_{\text{LMM}})_{\text{metal}}}{I(\text{Cu}_{2p3})_{\text{metal}}}$$

The metallic Cu and Cu_2O cannot be distinguished using Cu 2p3, whereas metallic Cu and Cu_2O could be distinguished from Cu LMM Auger spectra (Table 1). CuO was characterized by shake-up satellites with binding energies higher than that of the main Cu 2p3 and 2p1 peaks of about 10 eV.²⁵⁻²⁷ Thus, the binding energy of Cu 2p3 and the kinetic energy of Cu LMM were used in UHV and air before and after autoxidation to better analyze the Cu oxide layers.

2.1.2. Preparation of Cu sheet

Cu sheet purchased from Thermo Fisher Scientific with less than 0.01 purity content served as the sample. This sample was cleaned by sputtering for about 2 h with Ar ion gun (with 3 kV beam energy, 3 μA ion current density, and 3 mm $\times$ 3 mm sputtering area) until the peaks of C 1s and O 1s disappeared, and the pressure of analysis chamber was 1×10^{-8} mbar. The Cu sheet was then kept in UHV and air at room temperature, wherein the relative humidity of the air was about 20%.

2.2 Computational methods and models

DFT calculations were performed using the Dmol³ Materials Studio software package.^{28,29} The electronic structures were obtained by solving the Kohn–Sham equation self-consistently under spin-unrestricted conditions.^{30,31} DFT was also used for core electrons by applying the PW91 generalized-gradient approximation to the exchange-correlation energy.³² A double-numeric quality basis set with polarization functions was used. The transition state (TS) was identified using the complete linear/quadratic synchronous transit method.³³

X-ray diffraction characterization proved that the Cu(111), Cu(100), and Cu(110) surfaces, and the Cu_2O (111), Cu_2O (100), and Cu_2O (110) surfaces were the main surfaces of the metallic Cu and Cu_2O ,^{34,35} hence, H_2O and O_2 adsorption on these surfaces were studied. For the Cu(hkl) surfaces, surfaces were modeled using a four-layered mode $p(4 \times 4)$ super cell. For the Cu_2O (111),

Cu₂O(110) and Cu₂O(100) surfaces, each surface was modeled using a twelve-, six-, and eight-layered mode p(2 × 2) super cell, respectively. A 1.5 nm vacuum slab was used to separate the periodically repeated slabs. In all calculations, the bottom two layer of the Cu(hkl), Cu₂O(110), and Cu₂O(100) surfaces were fixed, and the bottom three layers of the Cu₂O(111) surface were fixed, whereas the other layers with the adsorbates were allowed to relax whereas the volume was maintained constant.

The rate constants of O₂ and H₂O dissociation can be described as follows:^{36,37}

$$k = A \exp\left(-\frac{E_a}{k_B T}\right) = \frac{k_B T}{h} \frac{Q_{TS}}{Q_R} \exp\left(-\frac{E_a}{k_B T}\right)$$

where h , k_B , A , T , and E_a are the Planck constant, Boltzmann constant, prefactor, reaction temperature (298 K), and activation energy, respectively. Q_{TS} and Q_R are the partition functions per unit volume for a transition state and a reactant, respectively.

According to statistical thermodynamics, Q can be estimated by vibrational partition functions as follows:

$$Q^{vib} = \prod_i \frac{\exp(-h\nu_i / 2k_B T)}{1 - \exp(-h\nu_i / k_B T)}$$

where ν_i is the frequency of vibrational mode i .

Hence, prefactors can be calculated by $A = \frac{k_B T}{h} \frac{Q_{TS}^{vib}}{Q_R^{vib}}$.

3. Results and discussion

The spectra of Cu sheet before autoxidation after 2 h sputtering is shown in Fig. 1. Only Cu peaks can be seen (Fig. 1a), and no O 1s and C 1s peaks can be observed (Figs. 1d and 1e), indicating that the sample is metallic Cu. The binding energy of Cu 2p3 and kinetic energy of Cu L₃VV are 932.7 and 918.1 eV, respectively (Figs. 1b and 1c), which agrees with previous results.¹³

After peak fitting, the Cu LMM peak of metallic Cu is found to be composed of five peaks (Fig. 1c), and the kinetic energy from low to high are as follows: 913.5 (0), 916.2 (1), 918.1(2), 919.3 (3) and 920.9 (4) eV (Table S1). Platzman et al.¹² also proposed that the Cu LMM is composed of five peaks but considered that the peak of about 916.2 eV can be attributed only to Cu₂O. According to the present results, the peak at 916.2 eV belongs to metallic Cu because there is no O 1s peak (Fig.1), and the area ratio of peaks 1 to 2 is 0.48. Given that peak 1 is entirely attributed to Cu₂O,

the content of Cu₂O obviously increases. Therefore, the relative area of peak 1 is deducted when the peak ratio of Cu and Cu₂O is calculated. In addition, given that the peak of Cu LMM is fitted, the kinetic energy and full width at half-maximum (FWHM) are considered.

3.1 Autoxidation process of Cu sheet in UHV

The peaks of Cu 2p and Cu LMM during the time of exposure to UHV are shown in Fig. S1. The binding energies of Cu 2p_{3/2} show no significant change with increased exposure time. The absence of shake-up satellite peak at about 943 eV indicates that no Cu²⁺ species are found after 1226 h of exposure to UHV. The relative intensities of Cu LMM between 917 and 924 eV also show no significant change, but the relative intensities of Cu LMM between 910 and 917 eV slightly increase with increased exposure time. The results of Cu LMM peak fitting are shown in Table S1, and the corresponding figure is shown in Fig. S2. The area ratio (peaks 1 and 2) after 1226 h of exposure to UHV is 0.52:1 compared with the Cu LMM area ratio (peaks 1 and 2) of metallic Cu, which is 0.48:1 (Table S1). This result shows that the area ratio of Cu₂O/Cu is about 0.04:1 (Table 2).

Table 3 shows the peak fitting result of O 1s exposed to UHV for 129, 183, 398, and 1226 h, but only the figure of the spectra exposed to UHV for 1226 h is shown (Fig. 2). As shown in Table 3 and Fig. 2, the O 1s spectra are composed of two peaks. The first peak at a low binding energy can be assigned to the O of Cu₂O or from dissociated adsorption O₂; the second peak at high binding energy can be attributed to OH from the dissociated adsorption H₂O.^{12,38,39} The mole ratio of O is found to be larger than that of OH, whereas the relative content of OH increases with increased exposure time (Table 3).

The change in Cu/O mole ratio continuously grows as a function of exposure time to UHV (Fig. 3). Fig. 3 shows that the Cu/O mole ratio during the first 129 h significantly decreases from 42.96 to 16.24. Afterwards, the Cu/O mole ratio slightly increases from 16.24 to 22.15 between 129 and 169 h, decreases from 22.15 to 7.34 between 169 and 348 h, gradually decreases between 348 and 746 h, and finally stabilizes. Three behaviors are found when O₂ and H₂O are present above the metal: dissociated adsorption, diffusion into bulk, or oxidation. The autoxidation of Cu sheet in UHV is found to be very slow during 1226 h, whereas the area ratio of Cu₂O/Cu is only 0.04 based on the above results; hence, the autoxidation of Cu sheet in UHV is not considered.

Therefore, the rate of O₂ and H₂O dissociated adsorption is far faster than of OH and O diffusion during the first 129 h. The rate of OH and O diffusion then is larger than that of O₂ and H₂O dissociated adsorption between 129 and 169 h. Afterwards, the rate of O₂ and H₂O dissociated adsorption is again faster than of OH and O diffusion until 746 h. Finally, the rate of OH and O diffusion becomes similar to of O₂ and H₂O dissociated adsorption. In addition, the peak of adsorption H₂O is not observed around 533 eV,^{12,40} indicating that H₂O undergoes dissociated adsorption in UHV. The mole ratio of O/OH (Table 3) indicates that the rate of O₂ dissociated adsorption is faster than that of H₂O dissociated adsorption.

3.2 Autoxidation process of Cu sheet in air

The peaks of Cu 2p and Cu LMM during the different times of exposure to air are shown in Figs. S3 and S4. The absence of shake-up satellites of the Cu 2p spectra when the exposure time is 500 h indicates that no formation of CuO (Fig. S3). The result agrees with previous ones wherein the formation of CuO is very slow at low temperature.^{13,14} However, compared with the Cu LMM of metallic Cu, the peak intensity at about 916.1 eV of Cu₂O obviously increases, indicating the autoxidation of metallic Cu and the formation of Cu₂O on the surface. The peak fitting results of Cu LMM and Cu 2p₃ are shown in Tables 2, S1, and S2, as well as Figs. 4 and 5. The Cu₂O contents are found to obviously increase with increased exposure time (Figs. 4 and 5). When the exposure times are 10 min and 1 h, the area ratios of Cu₂O/Cu are 0.37 and 0.62. Compared with the area ratio of Cu₂O/Cu after 1226 h (0.04) of exposure to UHV, the result shows that the increased exposure pressure accelerates the autoxidation degree. Therefore, a thin Cu₂O oxide layer easily forms on the surface in air.

When the exposure time is about 700 h, Cu 2p₃ is found to be obviously asymmetrical and a shoulder peak occurs; however, a shake-up satellite is not found (Fig. 5a). Iijima et al.¹³ also found that the shake-up satellites is obscure when the take-off angle is 90°, but the shake-up satellite intensity of Cu²⁺ species obviously increases with decreased take-off angle. The present result shows that Cu²⁺ species are formed. After peak fitting of Cu 2p₃, three peaks at 932.7, 933.7, and 935.1 eV are formed, which can be attributed to Cu(Cu₂O), CuO, and Cu(OH)₂, respectively.^{12,41,42} The corresponding Cu LMM peak is composed of five peaks (Fig. 4c). According to previous results, the peak at about 920.8 eV is found to be a unique peak of metallic Cu.⁴¹ The present

results show that the metallic Cu, Cu⁺, and Cu²⁺ layers are co-existent, in agreement with the finding of Platzman et al.¹³ Therefore, the CuO layer is formed even when the autoxidation of the Cu layer is not yet complete.

When the exposure time is about 900 h, the shake-up satellite appears (Fig. 5b), indicating that the content of Cu²⁺ species has increased. The corresponding Cu LMM peak is composed of only four peaks, and the peak at about 920.8 eV disappears (Fig. 4d). This result shows that the surface metallic Cu has totally completed oxidation and that the surface is composed of Cu₂O, CuO, and Cu(OH)₂ within the sample depth range of XPS. With increased exposure time, the intensity of the shake-up satellite increases (Figs. 5c and 5d), which shows that the content of Cu²⁺ species also increases (Table 2).

The O 1s XPS spectra (Fig. S5) and peak fitting results of the Cu sheet autoxidation in air are shown in Fig. 6 and Table 4. The peaks at 530.5 and 531.9 eV also can be assigned as O and OH. Compared with the O 1s spectra of the Cu sheet exposure to UHV, a new peak appears at 533.2 eV, which can be attributed to non-dissociated adsorption H₂O.^{12,40} The Cu/O mole ratio is found to obviously decrease with increased exposure time, which indicates that Cu sheet is covered by O species (Table 4). In addition, the relative content of O species decreases with increased exposure time, the relative content OH species increases, and the relative content H₂O slightly decreases.

3.3 Thickness of oxide species in air and UHV

According to the method in Section 2.3 and the data in Table 2, the thickness of Cu₂O is about 0.16 nm after exposure to UHV for 1226 h. The sample depth (*d*) is found to be equal to $3\lambda\sin\theta$.⁴³ Furthermore, the sample depth of Cu, Cu₂O, and CuO is about 4.05, 8.58, and 8.4 nm, respectively. Therefore, the thickness of Cu₂O layer of the total sample depth is about 3.95% after exposure to UHV for 1226 h.

The thickness of each oxide species (CuO and Cu₂O) in air at each exposure time is obtained from peak fitting the Cu LMM and Cu 2p₃ XPS spectra (Fig. 7). The thickness of Cu₂O is about 1.21 and 1.81 nm after exposure to air for 10 min and 1 h, indicating that the native oxidation of a Cu sheet into a Cu₂O layer started immediately after exposure to air. Furthermore, the thickness of the Cu₂O layer gradually increases with increased exposure time. When the exposure time is 700 h, the thickness of Cu, Cu₂O, and CuO layers in the sample depth of XPS is about 0.89, 5.32, and

1.31 nm, respectively. This result shows that the metallic Cu on the surface is at 78% autoxidation. Subsequently, the metallic Cu layer disappears because the sample depth is beyond the scope of XPS, and the thickness of CuO is about 2.13 nm when the exposure time is 900 h. Finally, after 1800 h of exposure, the thickness of the CuO layer is 4.02 nm. Given that Cu₂O layer gradually moves to the Cu bulk beyond the scope of XPS sample depth with increased exposure time, the thickness of Cu₂O layer cannot be determined; hence, the thickness of Cu₂O layer after 700 h of exposure is not given in Fig. 7. As shown in Fig. S6, the inverse-logarithmic growth rate law is observed, and the plot of the inverse oxide layer thickness serves as a function of the natural logarithm of exposure time.^{12,14}

After exposure to 1266 and 1800 h in UHV and air, the Cu sheets are sputtered using an Ar ion gun. The spectra of Cu LMM after 10 and 60 s sputtering are similar to that of metallic Cu, indicating that the Cu₂O or CuO layer is removed. The corresponding Cu/O mole ratio is 17.42 and 14.10, indicating that the surface has numerous O species (Table S3). According to the result of O 1s peak fitting (Fig. S7), two O species are found: O and OH (mole ratio $\approx$ 4 and 0.46; Table S3). This result shows that O and OH diffuse into the Cu bulk when Cu sheet is exposed to UHV and air. The relative content of O in bulk in UHV exposure is much larger than that of OH, but the relative content of O in bulk in air exposure is smaller than that of OH. This result shows that the diffusion rate of O is larger than that of OH in UHV, but the result is inversed with the presence of air. In addition, when the Cu₂O and CuO layers are removed using the etched Ar ion gun, a number of O species can be found on the Cu sheet, which indicates that the diffusion rate of O species is faster than the oxidation rate.

After Ar ion gun sputtering for 15 s, the peaks of Cu 2p_{3/2} at about 933.8 and 935.1 eV disappear and become symmetrical, and the shake-up peak of Cu 2p also disappears (not shown). Fig. 8 shows the Cu LMM spectra of Cu sheet exposed to 1800 h in air and etched as a function of sputtering time. The Cu LMM peak shows that the Cu sheet comprises Cu and Cu₂O, and the area ratio of Cu₂O/Cu is larger than that of the sample exposed to air for 1 h. After the Cu sheet is etched for 20 and 15 s, the Cu LMM spectrum is found to be similar to that of metallic Cu; only the Cu LMM spectrum intensity at low kinetic energy is slightly higher than that of metallic Cu. These results indicate that only a few Cu₂O particles exist on the Cu sheet, but the difference

cannot be observed with the naked eye unless compared with metallic Cu. Finally, the Cu LMM spectrum after 60 s sputtering is similar to that of metallic Cu. The depth etched result clearly shows that the main role of Ar ion gun is to clear the surface oxide layer in the first stage and then clear the diffusion O and OH into the bulk when XPS is calibrated by metallic Cu. Results also show that the diffusion rate is larger than the autooxidation rate independent of the difference in exposure pressure or time.

Based on the above results, the native oxide layer thickness depends on the exposure time, consistent with previous results under similar exposure conditions.^{12-14,18} Excluding the differences in thickness of the Cu₂O layer, three differences in the result are found compared with previous findings. The first difference is regarding the influence of pressure. Lampimäki¹⁸ showed that the influence of pressure of O₂ is from 3.7×10^{-2} mbar to 213 mbar, whereas the influence of pressure of O₂ is from about 4.2×10^{-10} mbar to 21 mbar in the present study. The second difference is regarding OH and O diffusion. Previous studies have not found OH or O diffusion because of the high exposure pressure,^{12,13,17-19} whereas present results indicate that the Cu₂O layer forms very rapidly in air, and OH and O diffusion is thus not obvious. However, autooxidation is very slow in UHV, whereas diffusion is obvious. The third difference is regarding the exposure time of the formation of CuO and Cu(OH)₂ layers. One reason for such difference lies in the overall composition, microstructure, and surface morphology of the lattice defects of the Cu thin film surface proposed by Platzman et al.¹² The other reason is the air humidity. Furthermore, Platzman et al. found that the area ratio of Cu(OH)₂ to CuO is about 1.4–1.7 after exposure to air for 24 h,¹² wherein the relative humidity is about 60%. Meanwhile, in the present study, the mole ratio of Cu(OH)₂ and CuO is about 1.0 after exposure to air for 700 h, wherein the relative humidity is about 20%. Yamamoto et al.⁴⁰ found that the Cu(111) surface is mainly covered with OH and H₂O under water vapor. Therefore, we believe that the dissociated adsorption of O₂ is easier than that of the dissociated adsorption H₂O over the Cu(hkl) surfaces in air at room temperature, and Cu₂O species is formed. Furthermore, the ability of O₂ dissociated adsorption is similar to or smaller than that of H₂O dissociated adsorption on the Cu₂O(hkl) surfaces. The relative ratio of H₂O in the air plays an important role for Cu₂O further oxidation. Given that the relative humidity of the air is large, Cu (OH)₂ is formed with content larger than that of CuO.

Moreover, given that the relative humidity of the air is small, $\text{Cu}(\text{OH})_2$ is formed with content smaller than that of CuO . A small content of $\text{Cu}(\text{OH})_2$ cannot be observed under the metastable $\text{Cu}(\text{OH})_2$, which undergoes thermal decomposition with ease in the range of 298 to 1273 K in air.^{44,45} To prove the hypothesis, the non-dissociated and dissociated adsorption O_2 and H_2O over the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces are studied using DFT (Section 3.4).

Therefore, the autoxidation of Cu sheet exposed to air is easier than that to UHV. Generally, the time of the sample preparation before ex-situ XPS characterization is approximately 10 min or even longer, and the valence or content of Cu species is altered by Cu autoxidation. As a result, the valence of Cu species and the relative content obtained by XPS yield inaccurate results. If the preservation time of the Cu-based catalysts is long or the preservation pressure is too high, the relative content of Cu^+ or Cu^{2+} may be greater. Therefore, we strongly recommend that the preservation time of the Cu-based catalysts be reduced or that the catalysts be preserved in a high vacuum and low relative humidity. Then, the Cu-based catalysts need sputter for approximately 10 s when it is detected by ex-situ XPS and other surface characterizations.

3.4. O_2 and H_2O adsorption over the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces

In this section, we investigate the O_2 and H_2O adsorption over the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces, and then describe the dissociation of H_2O and O_2 . Models of the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces are shown in Fig. S8. Four adsorption sites exist on the $\text{Cu}(111)$ surface: top, bridge (bri), face-centered cubic (fcc), and hexagonal close-packed (hcp). Four adsorption sites also exist on the $\text{Cu}(110)$ surface: top, long bridge (lb), short bridge (sb), and hollow (hol). Three adsorptive sites exist on the $\text{Cu}(100)$ surface: top, bridge (bri), and hollow (hol). For the $\text{Cu}_2\text{O}(\text{hkl})$ surfaces, the $\text{Cu}_2\text{O}(111)$ surface has hexagonal symmetry with four distinct atom types: Cu_{cus} , Cu_{csa} , O_{suf} , and O_{sub} . Based on the atomic coordinates, Cu_{cus} directly connects one O_{suf} atom that is coordinatively unsaturated; the surface O_{suf} connects three Cu atoms; Cu_{csa} is the coordinatively saturated copper atom that has the unique linear $\text{O}-\text{Cu}-\text{O}$ bond symmetry; O_{sub} is the subsurface oxygen that is fourfold coordinated. On the $\text{Cu}_2\text{O}(110)$ surface, only two adsorptive sites exist: Cu_{csa} and O_{suf} . On the $\text{Cu}_2\text{O}(100)$ surface, two adsorptive sites exist: Cu_{csa} and O_{suf} , and O_{suf} connects two Cu atoms. Figs. S9 and S10 show the adsorption configuration of O_2 , O, H_2O , OH, and H on the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces at their respective favorable sites. The

corresponding adsorption energies are listed in Table 5.

On the Cu(111) surface, the bond strengths of the parallel O₂ adsorption are stronger than that of the perpendicular O₂ adsorption (Table 5 and Fig. S9). This result agrees with previous studies.⁴⁶ However, the activation energy of the perpendicular O₂ adsorption is smaller than that of the parallel O₂ adsorption (see O₂ dissociation). Therefore, both preferred models are shown. The adsorption energies of the perpendicular and parallel O₂ adsorption on the fcc site are -0.28 and -0.97 eV, respectively. O, OH, and H are preferred to be adsorbed on the fcc sites, with adsorption energies of -4.53, -2.96, and -2.52 eV, respectively. The stable adsorption site of H₂O is the top site, and the corresponding adsorption energy is -0.18 eV. On the Cu(110) surface, the preferential adsorption configuration of the parallel O₂ adsorption is bound to the hollow site at which two O atoms bind with two adjacent long bridge sites, and the adsorption energy is -1.67 eV. O and H₂O are adsorbed on the pseudo-fcc and top sites, with adsorption energies of -4.93 and -0.65 eV, respectively. The stable configurations for the OH and H adsorbed at the short bridge site have adsorption energies of -3.46 and -2.57 eV. On the Cu(100) surface, the preferential adsorption configuration of O₂ is parallel adsorption at the hollow site at which two O atoms bind with two adjacent bridge sites, and the adsorption energy is -1.92 eV. The adsorption energy generated by H₂O adsorption on the Cu(100) surface is -0.24 eV. However, a specific adsorption site was not observed. O, OH, and H are adsorbed on the hollow site, with adsorption energies of -5.33, -3.25 and -2.45 eV, respectively.

On the Cu₂O(111) surface, the most stable adsorption structure of O₂ is parallel to the surface on which one O atom connects with the Cu_{cus} and the other O atom connects with two adjacent Cu_{csa} (defined as 3Cu site). The corresponding adsorption energy is -0.92 eV. O atom is preferred to be adsorbed on the 3Cu site, and the adsorption energy is -4.20 eV. The stable configurations for the H₂O and OH adsorbed at the Cu_{cus} site have adsorption energies of -0.93 and -3.23 eV. H tends to be adsorbed on the O_{suf} site, and the adsorption energy is -2.78 eV. On the Cu₂O(110) surface, the preferential adsorption configuration of O₂ is parallel adsorption on the 2Cu site (two adjacent Cu_{csa}), and the adsorption energy is -0.11 eV. O atom tends to be adsorbed on the

Cu_{csa}-O_{suf} site (defined as Cu-O site), and the adsorption energy of O atom is -1.96 eV. There is no a specific adsorption site for H₂O, and the adsorption energy is -0.42 eV. The stable configuration for OH adsorbed at the Cu_{csa} site has an adsorption energy of -1.62 eV. H tends to be adsorbed on the O_{suf} site, and the adsorption energy is -2.74 eV. On the Cu₂O(100) surface, O₂ and OH are preferred to be adsorbed on the 2Cu site, with adsorption energies of -0.11 and -2.26 eV. O atom tends to be adsorbed on the Cu-O site, and the adsorption energy is -2.75 eV. No specific adsorption site is available for H₂O, and the adsorption energy is -0.32 eV. The stable configuration for H adsorbed at the O_{suf} site has an adsorption energy of -3.62 eV.

The potential energy diagram for O₂ dissociation together with the TS and final state (FS) is shown in Fig. 9. For the Cu(111) surface, the bond length of O₂ is increased to 0.138 nm from 0.134 nm in the initial state (IS) when O₂ is perpendicular adsorption in the TS1-0 case. The bond length of O₂ is increased from 0.149 nm in the IS to 0.193 nm in the TS1-1 case when O₂ is parallel adsorption. The O-O bond scissions are exothermic with reaction energies of 1.79 (TS1-0) and 1.10 eV (TS1-1), and activation energies of 0.13 and 0.50 eV. Two O atoms are adsorbed on two adjacent fcc sites, at which the bond lengths of O-Cu are 0.187, 0.198, 0.199, and 0.189 nm in the FS. On the Cu(110) surface, the length of the O-O bond is 0.193 nm in the TS (TS1-2). The activation and reaction energies of O₂ dissociation are 0.22 and -1.21 eV. Two O atoms adsorb on hollow sites, at which the bond lengths of O-Cu are 0.211 and 0.209 nm in the FS. On the Cu(100) surface, the length of the C-O bond is 0.189 nm in the TS (TS1-3). The activation energy is 0.21 eV with a reaction energy of -1.75 eV. Two O atoms are adsorbed on the hollow sites, and the bond length of Cu-O is 0.203 nm in the FS. On the Cu₂O(111) surface, the bond length of O₂ is increased to 0.225 nm in the TS from 0.139 nm in the IS. The process needs to overcome an activation energy of 1.20 eV with an exothermicity of 0.49 eV (TS1-4). Two O atoms are adsorbed on the 3Cu sites, and the bond lengths of Cu-O are 0.187, 0.192, 0.181, and 0.205 nm in the FS. For the Cu₂O(110) surface, the bond length of O₂ is increased from 0.126 nm in the IS to 0.222 nm in the TS. The activation energy is 3.97 eV with a reaction energy of 3.16 eV (TS1-5). On the Cu₂O(100) surface, the bond length of O₂ is increased to 0.229 nm in the TS from 0.127 nm in the IS. The process needs to overcome an activation energy of 2.56 eV with an

endothermicity of 1.61 eV (TS1-6).

The potential energy diagram for H₂O dissociation together with the TS and FS is shown in Fig. 10. On the Cu(111) surface, one bond length of H–O is increased to 0.175 nm in the TS from 0.098 nm in the IS. H and OH are adsorbed on two adjacent fcc sites in the FS. The H–O bond scission is slightly endothermic with a reaction energy of 0.03 eV and an activation energy of 1.45 eV (TS2-0). In the case of Cu(110), one bond length of H–O is increased to 0.170 nm in the TS from 0.097 nm in the IS. The process needs to overcome an activation energy of 1.10 eV with an exothermicity of 0.19 eV (TS2-1). On the Cu(100) surface, one bond length of H–O is increased from 0.097 nm in the IS to 0.142 nm in the TS. The activation and reaction energies of H₂O dissociation are 1.28 and –0.24 eV (TS2-2), respectively. On the Cu₂O(111) surface, one bond length of H–O is increased from 0.098 nm in the IS to 0.134 nm in the TS. The process needs to overcome an activation energy of 1.10 eV with an endothermicity of 0.24 eV (TS2-3). On the Cu₂O(110) surface, when OH is adsorbed on the Cu_{csa} site and H is adsorbed on the O_{surf} site near the Cu_{csa}, H₂O is formed after optimization. The result indicates that OH and H cannot form from H₂O dissociation. On the Cu₂O(100) surface, one bond length of H–O is increased to 0.148 nm in the TS from 0.097 nm in the IS, and the activation and reaction energies of H₂O dissociation are 0.06 and –0.30 eV (TS2-4).

Comparing with the activation energies of O₂ and H₂O dissociation on the Cu(hkl) surface, the activation energy of O₂ dissociation on the Cu(hkl) surface is obviously smaller than that of H₂O dissociation. This result indicates that O₂ dissociation is easier than H₂O dissociation. However, our XPS result shows that the mole ratio of OH:O is approximately 0.11:1 exposed to UHV for 108 h, showing that H₂O dissociation also occurs on Cu sheet. Whether or not the pre-adsorbed oxygen from O₂ dissociation affects H₂O dissociation remains unclear. Therefore, we also calculate the H₂O dissociation on the pre-adsorbed oxygen Cu(hkl) surface.

The potential energy diagram of H₂O dissociation on the oxygen-covered Cu(hkl) surfaces together with the IS, TS, and FS is shown in Fig. 11. On the oxygen-covered Cu(111) surface, one H–O of H₂O points out toward the pre-adsorbed O atom in the IS. The O–H bond length is 0.099 nm, and the adsorption energy of H₂O on the oxygen-covered Cu(111) surface is –0.49 eV.

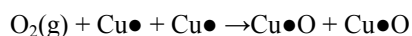
Comparing with H₂O adsorption on the clean Cu(111) surface, the adsorption energy (−0.49 vs. −0.18 eV) and O–H bond length (0.099 vs. 0.097 nm) show that the pre-adsorbed O atom will improve the stability of H₂O adsorption and the extent of H₂O activation. The bond length of the activated O–H is increased to 0.155 nm in the TS from 0.099 nm in the IS. The process needs to overcome an activation energy of 0.43 eV with an exothermicity of 0.39 eV (TS2-5). On the oxygen-covered Cu(110) surface, one H–O of H₂O also points out toward the pre-adsorbed O atom in the IS, and the bond length of O–H is 0.105 nm. The distance of O and H in the TS is 0.132 nm, and the activation and reaction energies of H₂O dissociation are 0.20 and −0.62 eV (TS2-6). On the oxygen-covered Cu(100) surface, one H–O of H₂O also points out toward the pre-adsorbed O atom in the IS. The bond length of the activated O–H is increased to 0.189 nm in the TS from 0.100 nm in the IS. The H–O bond scission is exothermic with a reaction energy of 0.29 eV and an activation energy of −0.13 eV (TS2-7). Comparing with the activation energies of H₂O dissociation on the clean and oxygen-covered Cu(hkl) surfaces, the activation energies of H₂O dissociation on the oxygen-covered Cu(hkl) surfaces is obviously smaller than those on the clean Cu(hkl) surfaces, although the activation energies of H₂O dissociation on the oxygen-covered Cu(hkl) surfaces are also slightly larger than that of O₂ dissociation on the clean Cu(hkl) surfaces. However, the difference is small. Thus, H₂O dissociation possibly occurs on the oxygen-covered Cu(hkl) surfaces.

The rate constants (*k*) for O₂ and H₂O dissociation on the Cu(hkl) and Cu₂O(hkl) surfaces at *T* = 298 K are listed in Table 6, where the prefactors and activation energies are also included. As shown in Table 6, the rate constants of O₂ dissociation on the clean Cu(hkl) surfaces are obviously larger than that of H₂O dissociation, indicating that O atom adsorption on the Cu(hkl) surfaces is easier than that of OH and H. Then, the Cu sheet is easily autooxidation when the pressure of O₂ is adequate. Therefore, the Cu₂O layer is formed after exposure to 10 min in air, but the Cu₂O layer is formed after exposure to approximately 1266 h in UHV. When the Cu surface is covered by O atoms, the rate constant of H₂O dissociation obviously increases compared with the rate constant of H₂O dissociation on the clean Cu surface. This result indicates that the pre-adsorbed O will accelerate H₂O dissociation. Yamamoto et al.⁴⁰ also found that only O occurs on the clean Cu(111)

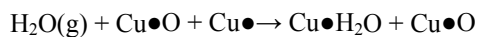
surface using XPS, but the OH and adsorption H₂O peaks appear on the oxygen-covered Cu(111) surface. In addition, the rate constants of H₂O dissociation on the oxygen-covered and clean Cu(110) surfaces are larger than those of H₂O dissociation on the oxygen-covered and clean Cu(111) surfaces. The result also agrees with previous results that the hydroxylation of the Cu(110) surface is facilitated by a lower activation barrier for H₂O dissociation compared with the Cu(111) surface.^{17, 40}

When the Cu₂O layer is formed, the rate constants of O₂ dissociation on the Cu₂O(hkl) surfaces are obviously smaller than those of O₂ dissociation on the Cu(hkl) surfaces. The result also agrees with the previous result that CuO layer formation is harder than Cu₂O layer formation.¹³ However, the rate constants of H₂O dissociation on the Cu₂O(hkl) surfaces are larger than those of O₂ dissociation. In other words, when the relative humidity of air is adequate, Cu(OH)₂ layer formation may be easier than CuO layer formation. This result agrees with the result of Platzman et al..¹²

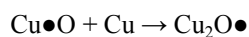
Basing on our results obtained by XPS and DFT combined with previous literature, we determine that the autoxidation mechanism on the Cu sheet is as follows: (a) The dissociative adsorption of O₂, as described by the following reaction:



where Cu• and Cu•O indicate the adsorption sites of metallic Cu and metallic Cu adsorption O. The dissociative rate depends on the exposure pressure, which accelerates the O₂ dissociative adsorption. (b) The coexistence of non-dissociative and dissociative H₂O adsorption can be described by the following reaction:

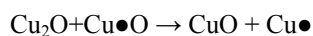
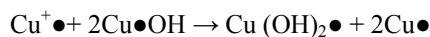


This stage is induced by the adsorption O and also depends on the relative humidity and its pressure. (c) In the diffusion of O and OH into Cu bulk, the rate is significantly greater than the autoxidation rate. (d) The Cu₂O layer formation, known as a p-type semiconductor,⁴⁷ can be described by the following reaction:



This stage is induced by an electric field (as a driving force at low-temperature conditions)

formed by positive ions of Cu at the metal/oxide interface and negative ions at the oxide/gas interface.^{12,48, 49} The stage initially depends on the concentration of the adsorbed oxygen on the surface. The growth of the Cu₂O layer at this stage is affected by the microstructure and morphology of the surface and by the content of lattice defects that will induce the metal cations to migrate.^{12,50} (e) The metastable Cu(OH)₂ and CuO layer formation can be described by the following reaction:⁵¹



The Cu(OH)₂ layer formation is affected by the concentration of adsorbed OH, which reacts with the Cu⁺ of the surface. (f) The transformation phase of the metastable Cu(OH)₂ to CuO can be described by the following reaction:



The transformation phase rate of this step depends on the relative humidity of the air. Previous studies found that the transformation phase rate is very fast at room temperature in aqueous surroundings.^{44,52}

4. Conclusion

This study investigates the autoxidation mechanism of Cu sheet through XPS and DFT. Basing on the peak fitting of Cu 2p3 and Cu LMM, we determine that the Cu sheet is not easily exposed to UHV for 1226 h and that the thickness of the Cu₂O layer is approximately 0.16 nm. The thickness of Cu₂O is approximately 1.21 and 1.81 nm after exposure to air for 10 min and 1 h. When the exposure time is 700 h, the CuO layer formed is approximately 1.31 nm thick. The metallic Cu of the Cu sheet surface is total autoxidation after exposure to 900 h. Finally, the thickness of the CuO layer is 4.02 nm after 1800 h exposure.

Basing on the results of XPS and DFT analyses, we propose that the autoxidation mechanism consists of six stages. First is the dissociative adsorption of O₂, wherein the dissociative rate depends on the exposure pressure, which accelerates O₂ dissociative adsorption. Second is the coexistence of non-dissociative and dissociative H₂O adsorption. This stage is induced by the adsorption O and depends on the relative humidity and its pressure. Third is the diffusion of O and OH into Cu bulk, wherein the rate is far greater than that of the autoxidation rate. Fourth is Cu₂O

layer formation induced by an electric field formed. Fifth is the metastable $\text{Cu}(\text{OH})_2$ and CuO layer formation. The relative content of $\text{Cu}(\text{OH})_2$ depends on the relative humidity of air, and it increases with increased relative humidity. Sixth is the transformation phase of the metastable $\text{Cu}(\text{OH})_2$ to CuO . The formation of $\text{Cu}(\text{OH})_2$ depends on the relative humidity of air. The $\text{Cu}(\text{OH})_2$ phase may not be observed when the relative humidity is too small.

For Cu-based catalysts, the preservation time of the Cu-based catalysts should be reduced or the catalysts should be preserved in a high vacuum and low relative humidity. Considering the time of the sample preparation before ex-situ XPS and other surface characterizations, we recommend that the Cu-based catalysts be sputtered for approximately 10 s using Ar ion gun.

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Supporting Information

Detailed information on the Cu 2p, Cu LMM and O 1s spectra during the different times of exposure to UHV and air, dependence of the inverse oxide layer thickness on the natural logarithm of exposure time to air, O 1s spectra of the Cu sheet after Ar ion gun etched, structure and adsorption sites of the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces, most stable configurations of adsorption species on the $\text{Cu}(\text{hkl})$ and $\text{Cu}_2\text{O}(\text{hkl})$ surfaces. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Figure captions:

Fig. 1 XPS of the Cu sheet after sputtering for about 2 h with Ar ion gun. (a), (b), (c), (d), and (e): survey spectrum, Cu 2p, Cu LMM, O 1s and C 1s peak.

Fig. 2 O 1s spectra of the Cu sheet exposed to UHV for 1226 h

Fig. 3 Mol ratio of Cu/O on the Cu sheet at different exposure times under UHV

Fig. 4 Cu LMM spectra of the Cu sheet measured at different exposure times in air. (a), (b), (c) and (d): 10 min, 300 h, 700 h, and 900 h.

Fig. 5 Cu 2p spectra of the Cu sheet measured at different exposure times in air. (a), (b), (c) and (d): 700, 900, 1300, and 1800 h.

Fig. 6 O 1s spectra of the Cu sheet measured at different exposure time in air. (a), (b), (c) and (d): 10 min, 72 h, 300 h, and 1800 h .

Fig. 7 Oxide layer thickness grown on the Cu sheet surface at different exposure times in air

Fig. 8 Cu LMM spectra of Cu sheet exposure to 1800 h in air after different etched time.

Fig. 9. Potential energy diagrams and the corresponding TS and FS for O₂ dissociation. TS1-0, TS1-1, TS1-2, TS1-3 TS1-4, TS1-5, and TS1-6: perpendicular adsorption on the Cu(111) surface and parallel adsorption on the Cu(111), Cu(110), Cu(100), Cu₂O(111), Cu₂O(110), and Cu₂O(100) surfaces. Bond lengths are in nm. Bond lengths are in nm. O, H, and Cu atoms are shown in the red, white, and orange balls, respectively.

Fig. 10. Potential energy diagrams and the corresponding TS and FS for H₂O dissociation. TS2-0, TS2-1, TS2-2, TS2-3, and TS2-4: on the Cu(111), Cu(110), Cu(100), Cu₂O(111), and Cu₂O(100) surfaces. Bond lengths are in nm. See Fig. 9 for colour coding.

Fig. 11. Potential energy diagrams and the corresponding IS, TS, and FS for H₂O dissociation on the pre-adsorbed oxygen Cu(111) (TS2-5), Cu(110) (TS2-6), and Cu(100) (TS2-7) surfaces. Bond lengths are in nm. See Fig. 9 for colour coding.

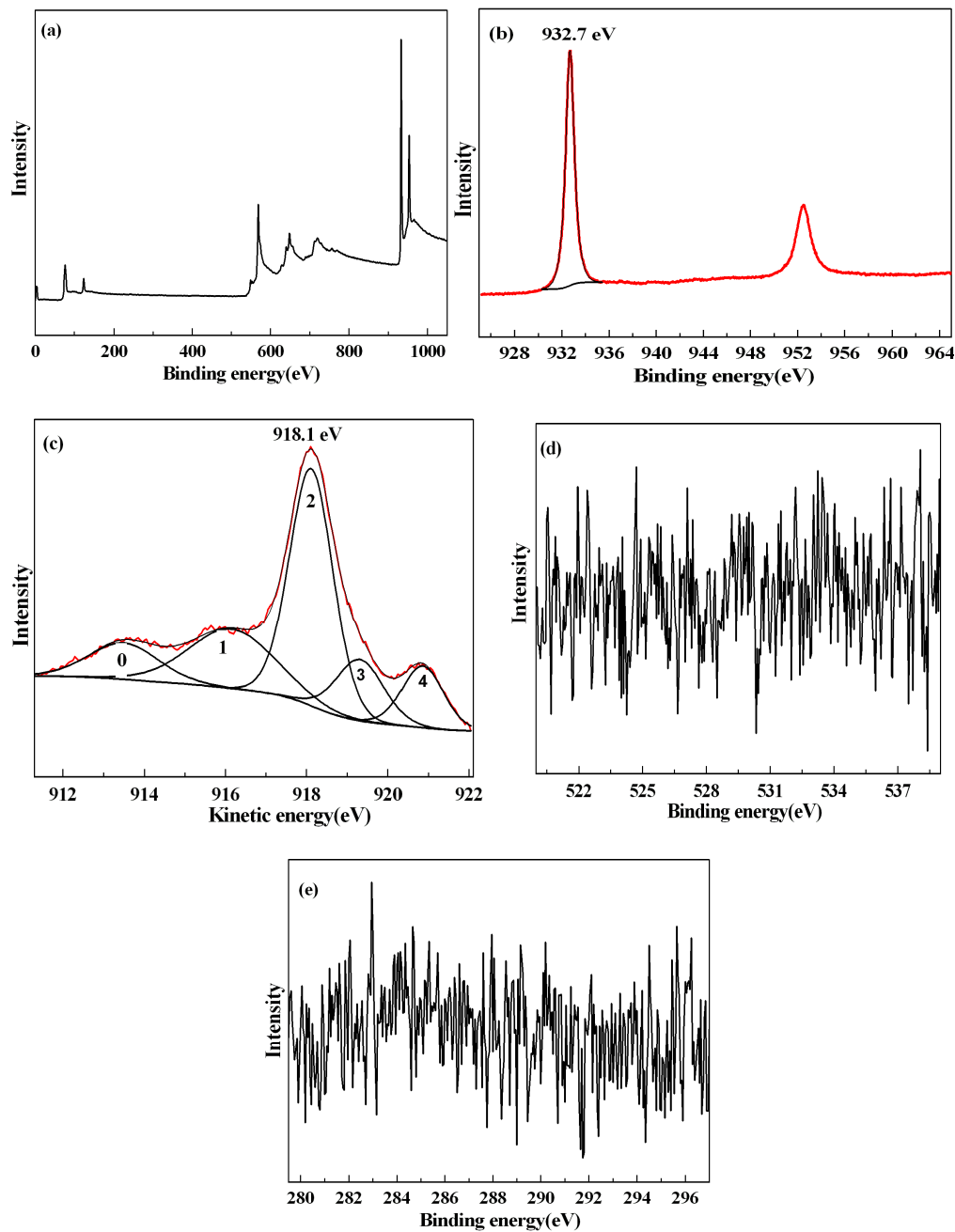


Fig. 1

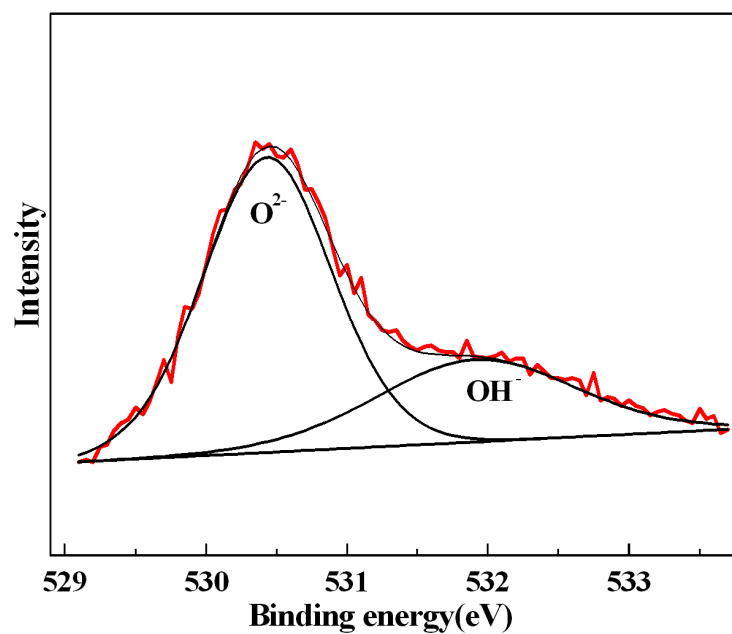


Fig. 2

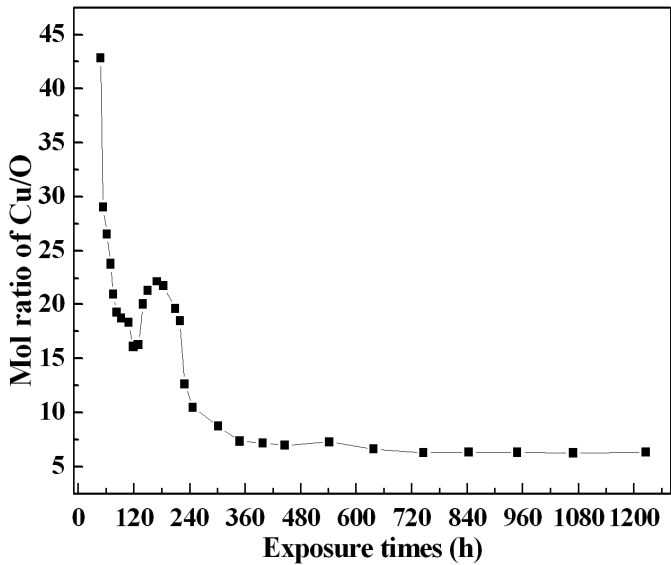


Fig. 3.

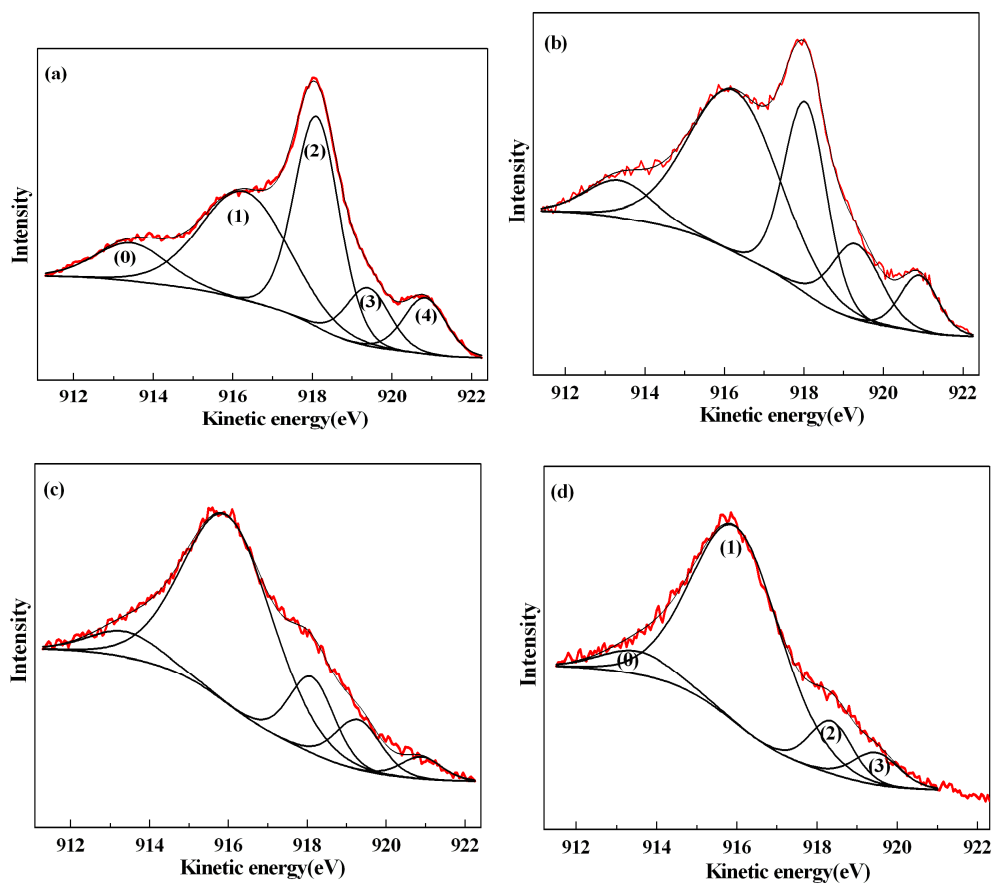


Fig. 4

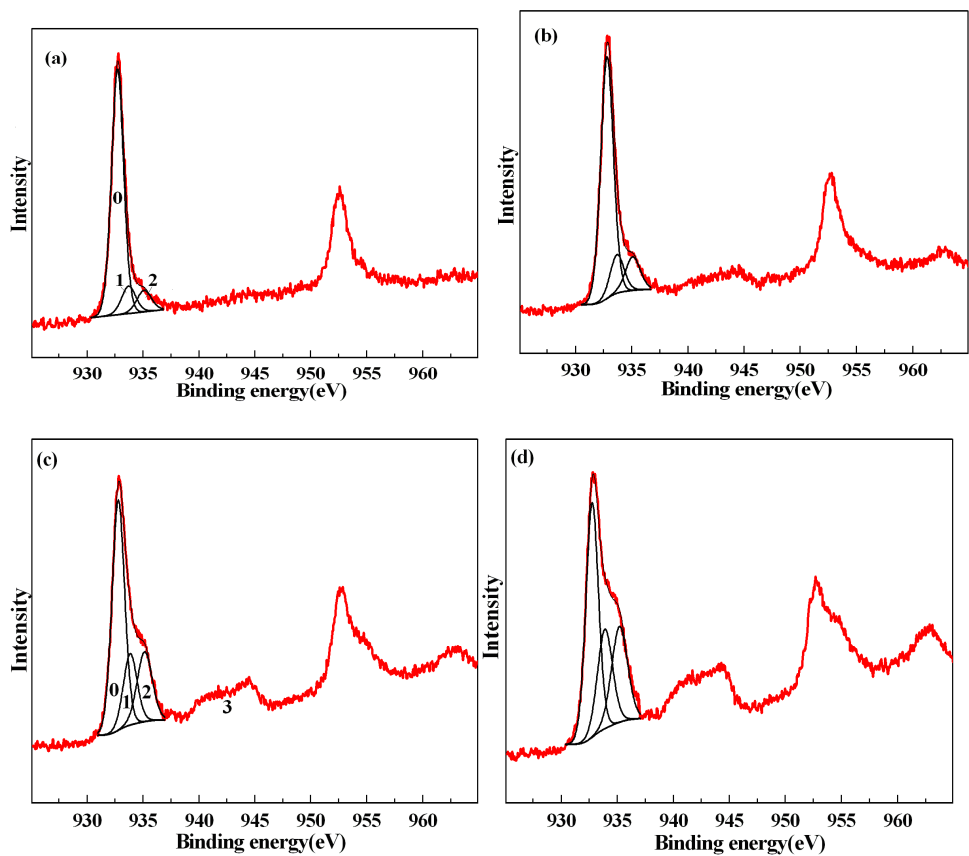


Fig. 5

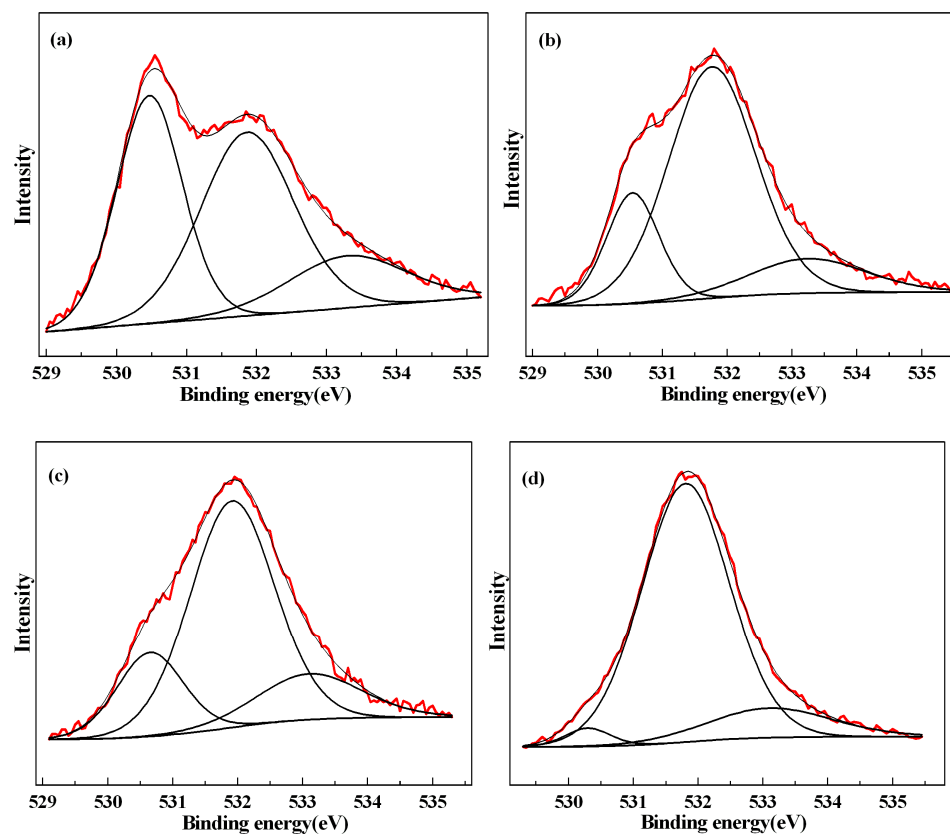


Fig. 6

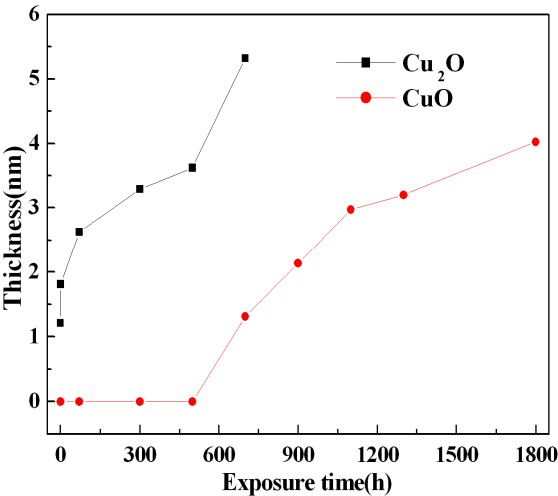


Fig.7

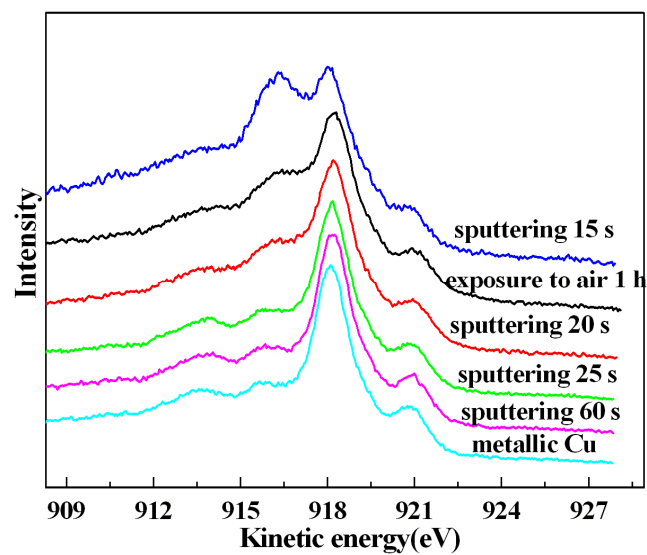


Fig.8

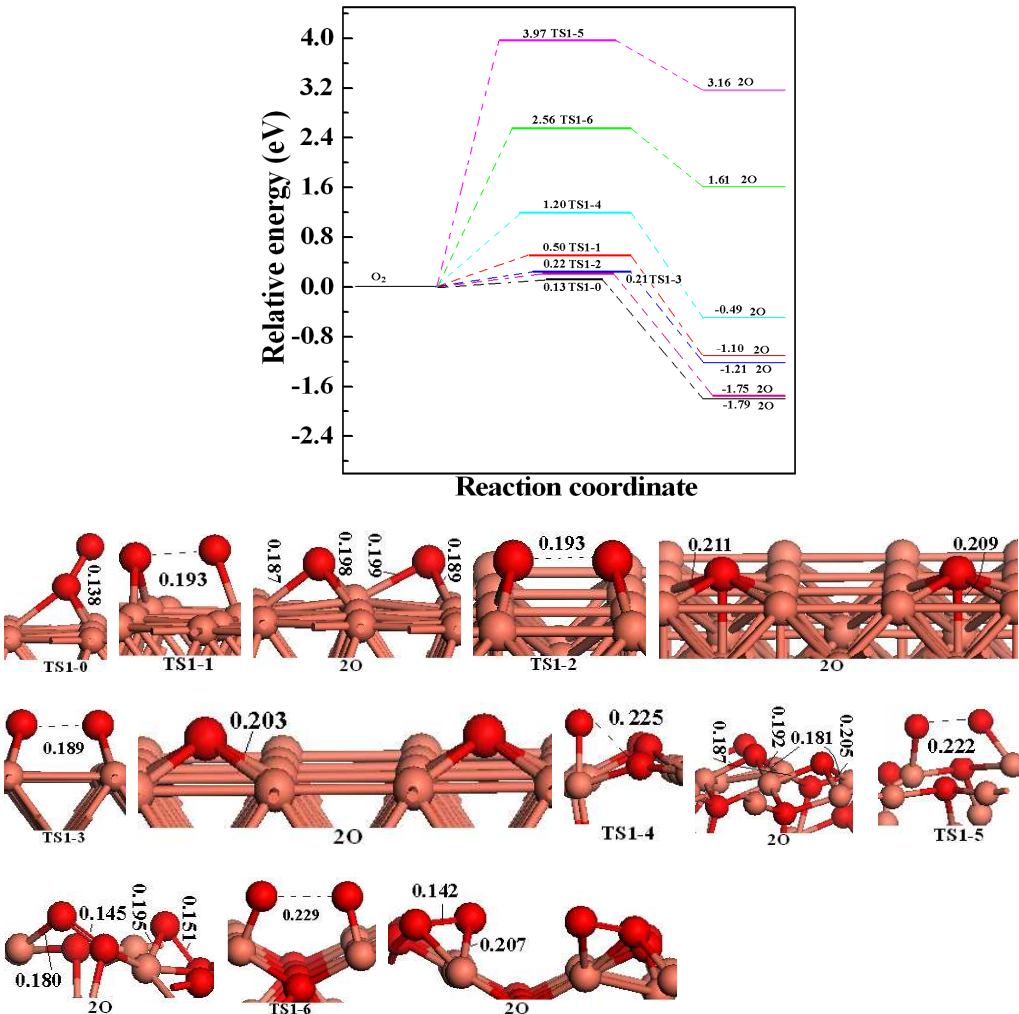


Fig. 9

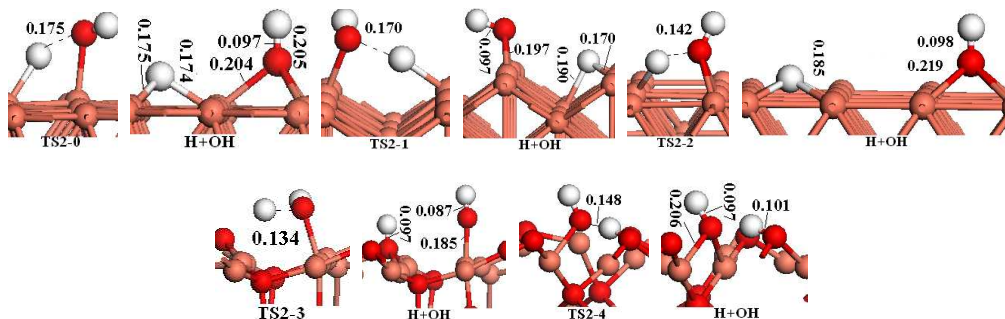
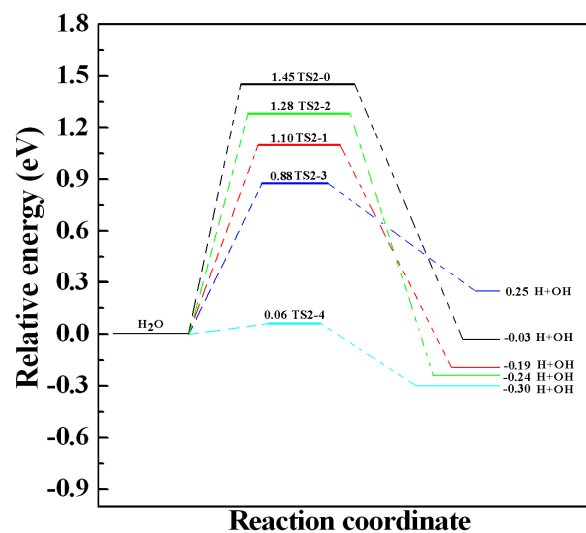


Fig. 10

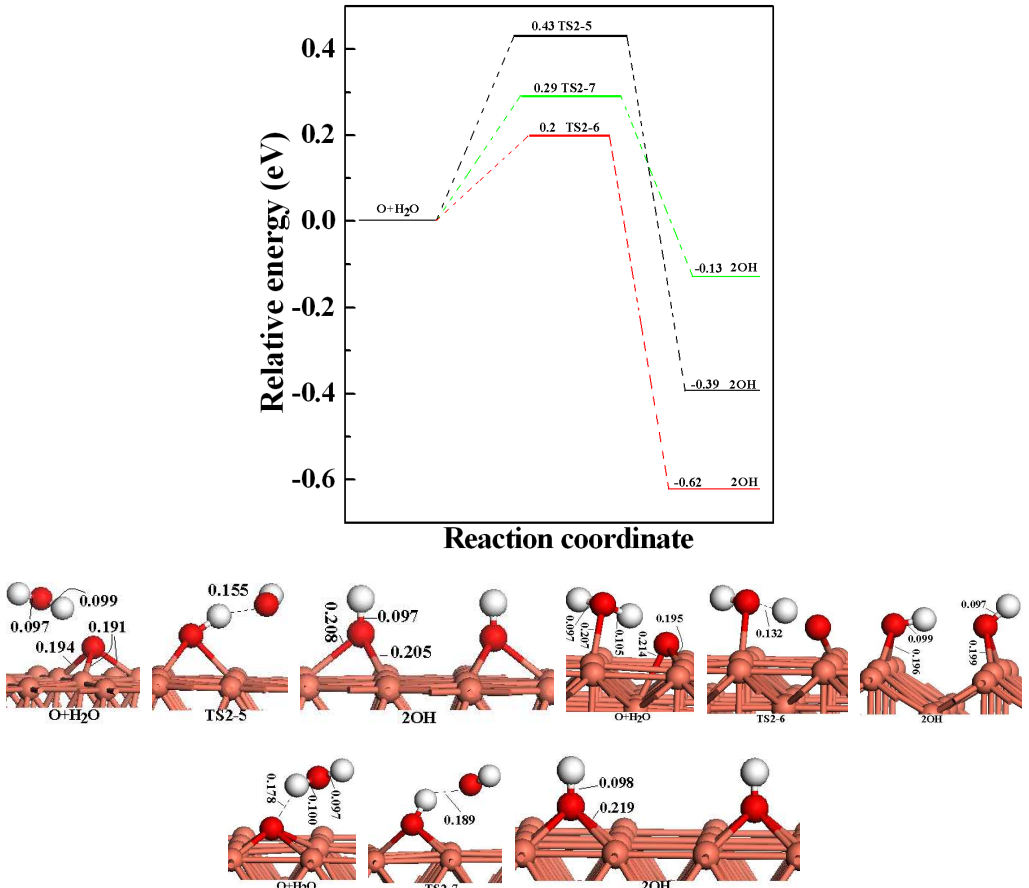


Fig. 11

Table 1 The positions of Cu L₃VV, Cu 2p₃ and O1s for copper-containing reference materials
using a X-ray Al K α .

material	kinetic energy (eV)	binding energy (eV)		reference
	Cu L ₃ VV	Cu 2p ₃	O 1s	
metallic Cu	918.4	932.6		13
Cu ₂ O	916.2	932.5	530.6	12
CuO	918.1	933.7	530.6	12
Cu(OH) ₂	916.2	935.1	531.8	12
H ₂ O adsorption on Cu			533.0	12

Table 2 Summary of the compositions and its area ratio (S) on the Cu sheet at different exposure times in UHV and air at room temperature

pressure	exposure time	material	S(Cu):S(Cu ₂ O):S(CuO)
UHV	0	Cu	1:0:0 ^a
	1226 h	Cu, Cu ₂ O	1:0.04:0 ^a
	10 min	Cu, Cu ₂ O	1:0.37:0 ^a
	1h	Cu, Cu ₂ O	1:0.62:0 ^a
	72 h	Cu, Cu ₂ O	1:1.06:0 ^a
air	300 h	Cu, Cu ₂ O	1:1.52:0 ^a
	500 h	Cu, Cu ₂ O	1:1.79:0 ^a
	700 h	Cu, Cu ₂ O, CuO(Cu(OH) ₂)	1:4.52: 0.25 ^b
	900 h	Cu ₂ O, CuO(Cu(OH) ₂)	0:1:0.6 ^c
	1100 h	Cu ₂ O, CuO(Cu(OH) ₂)	0:1:0.99 ^c
	1300 h	Cu ₂ O, CuO(Cu(OH) ₂)	0:1:1.12 ^c
	1800 h	Cu ₂ O, CuO(Cu(OH) ₂)	0:1:1.68 ^c

^a and ^c the values from the peak fitting results of the Cu LMM and Cu 2p3; ^b the values of 1:4.52 and 4.52: 0.25 from the peak fitting results of the Cu LMM and Cu 2p3.

Table 3 Summary of the peak fitting results of O 1s and mole ratio (M) at different exposure times
in UHV

exposure time	binding energy (eV)		M(OH):M(O)
	OH	O	
129 h	531.7(1.6) ^a	530.5(1.2)	0.11/1
183 h	531.7(1.7)	530.4(1.1)	0.19/1
398 h	531.8 (1.6)	530.4(1.0)	0.27/1
1266 h	531.9 (1.7)	530.4(1.1)	0.40/1

^a Corresponding FWHM of the peak.

Table 4 Summary of the peak fitting results of O 1s and mole ratio at different exposure times in air

exposure	binding energy(eV)			M(O):M(OH):M(H ₂ O _{ad}) ^a	M(Cu):M(O)
time	O	OH	H ₂ O _{ad}		
10min	530.5(1.2)	531.9(1.6)	533.3(2.1)	0.88/1/0.36	70.58/29.42
72 h	530.5(1.0)	531.8(1.6)	533.2(2.2)	0.29/1/0.20	61.84/38.16
300 h	530.6(1.3)	531.9(1.6)	533.1(2.0)	0.29/1/0.25	54.55/45.56
1800 h	530.3(0.9)	531.9(1.7)	533.1(2.2)	0.04/1/0.16	27.23/72.77

^a H₂O_{ad}: adsorption H₂O

Table 5: Adsorption energies (E_{ads} , eV) and adsorption configurations of reaction intermediates

surfaces	O ₂	config. ^a	O	config.	H ₂ O	config.	OH	config.	H	config.
Cu(111)	-0.28	fcc ^b	-4.53	fcc	-0.18	O-top	-2.96	O-fcc	-2.52	fcc
	-0.97	hcp								
Cu(110)	-1.67	hol	-4.93	pseudo-fcc	-0.65	O-top	-3.46	O-sb	-2.57	sb
Cu(100)	-1.92	hol	-5.33	hol	-0.24	no bound	-3.25	O-H	-2.45	hol
Cu ₂ O(111)	-0.92	3Cu	-4.20	3Cu	-0.93	O-Cu _{cus}	-3.23	O-Cu _{cus}	-2.78	O _{suf}
Cu ₂ O(110)	-0.11	2Cu	-1.96	Cu-O	-0.42	no bound	-1.62	O-Cu _{csa}	-2.74	O _{suf}
Cu ₂ O(100)	-0.11	2Cu	-2.75	Cu-O	-0.32	no bound	-2.26	O-2Cu	-3.62	O _{suf}

^a config.: configuration; ^b perpendicular adsorption of O₂

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Table 6. Activation energies (eV), prefactors A (s⁻¹), and rate constant k(s⁻¹) calculated at 298 K and activation energies (E_a) of O₂ and H₂O dissociation on the Cu(hkl) and Cu₂O(hkl) surfaces

reactions	surfaces	E _a	A	k
O ₂ →O+O	Cu(111)	0.13 ^a	3.64×10 ¹⁴	2.32×10 ¹²
		0.50	2.13×10 ¹⁴	7.64×10 ⁵
	Cu(110)	0.22	8.06×10 ¹⁴	1.55×10 ¹¹
	Cu(100)	0.21	1.98×10 ¹⁵	5.63×10 ¹¹
	Cu ₂ O(111)	1.20	1.71×10 ¹³	9.26×10 ⁻⁸
	Cu ₂ O(110)	3.97	1.39×10 ¹⁵	1.24×10 ⁻⁵²
	Cu ₂ O(100)	2.56	3.49×10 ¹⁴	2.03×10 ⁻²⁹
H ₂ O→OH+H	Cu(111)	1.45	1.82×10 ¹³	5.89×10 ⁻¹²
	Cu(110)	1.10	7.15×10 ¹⁴	1.89×10 ⁻⁴
	Cu(100)	1.28	5.13×10 ¹⁰	1.24×10 ⁻¹¹
	O-Cu(111) ^b	0.43	5.56×10 ¹³	3.04×10 ⁶
	O-Cu(110) ^b	0.20	7.19×10 ¹⁴	3.01×10 ¹¹
	O-Cu(100) ^b	0.29	5.84×10 ¹²	7.38×10 ⁷
	Cu ₂ O(111)	0.88	1.26×10 ¹³	1.73×10 ⁻²
	Cu ₂ O(110)	---	---	---
	Cu ₂ O(100)	0.06	2.65×10 ¹²	2. 57×10 ¹¹

^a perpendicular adsorption of O₂; ^b oxygen-covered Cu(111) surface